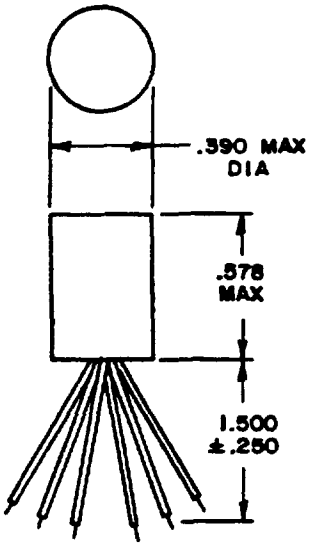
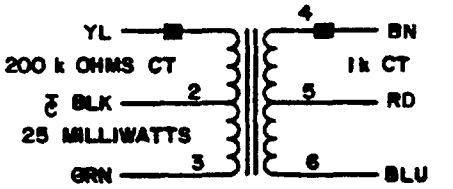
MILITARY SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY, TF5R21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.





WORKING VOLTAGE : 50 V
ALTITUDE: 150,000 FT
CIRCUIT DIAGRAM AND MARKING

INCHES	MM
250	6 35
.390	9.91
.578	14.68
1 500	38.10

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 inch = 25 4 mm.
3 Marking shall be on the side of the case

FIGURE 1. Dimensions and configurations

FSC 5950

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REQUIREMENTS:

Electrical ratings·
Primary power level 25 milliwatts maximum.
Working voltage. 50 volts peak.
Primary impedance (1-3) 200 K ohms CT.
Secondary load impedance (4-6) 1 K ohm CT.
Frequency range 300 to 20 kHz ±3 dB at 1 mW level.

Design and construction
Dimensions and configuration· See figure 1.
Duty cycle Continuous.
Case Metal.
Terminals Gold plated dumet, insulated wire in accordance with MIL-STD-1276, type D.
Diameter .016 inch.
Length· 1.500 ±.250 inch.
Weight 0.2 ounce maximum.
Operating temperature range· -55° to +105°C.
Altitude: 150,000 feet.

Thermal shock· MIL-STD-202, method 107, test condition A-1.

Terminal strength MIL-STD-202, method 211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (each winding)
At sea level 100 volts rms.
At reduced barometric pressure 100 volts rms.

Electrical characteristics
Primary impedance (1-3) 200 K ±20 percent ohms with approximately 10 volts, 5 kHz and 0 amp dc applied to (1-3) and 1 K ohms across (4-6).
Harmonic distortion. Total harmonic content of output 5 percent with 70 volts, 1 kHz and 0 ampere dc applied to (1-3) maximum.
Frequency response (Load voltage) Z_S = 200 k ohms (1-3), Z_1 = 1 K ohms (4-6), E_L = 1.0 volts, reference frequency = 1 kHz. Frequency range is 300 to 20 kHz ±3 dB.
Polarity Additive, with terminals (YEL) and (BLU) connected.
Insertion loss· ±3 dB with 70 volts, 1 kHz across (1-3) maximum.

Vibration (high frequency) MIL-STD-202, method 204, test condition B.

Shock (Specified pulse) MIL-STD-202, method 213, test condition H.

Marking location See figure 1.

Part number M27/263-01

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR, WC, ME
Navy - MC, AS
Air Force - 19

Preparing activity·
Army - ER

Agent
DLA - ES

(Project 5950-0561-4)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE MIL-T-27/263 TRANSFORMER, AUDIO FREQUENCY, TF5R21ZZ

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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